COLLEGE OF ENGINEERING, PUNE

(An Autonomous Institute of Government of Maharashtra.)

T2 Examination

B

Programme: B.Tech | **Semester: IV**

Course Code: - | **Course Name: Data Structures II**

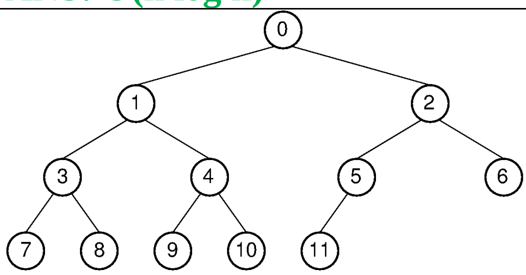
Branch: Computer Engineering | **Academic Year: 2021 -2022**

Duration: 20 mins | **Max Marks: 10**

Student PRN No.

Instructions:

1. Over writing is not allowed. Such answers will not be checked.
2. No marks will be awarded for partially correct answers.

1.	Prototype of swap function is: `void swap(int *a, int *b);` Write a function call statement to swap ith and jth element of a heap implemented using Array from within the `insert()` function **ANS:** **swap(&h->A[i], &h->A[j]);**	1/2
2.	2, 1, 3, 4 and 5 were inserted in an AVL tree. How many nodes will get imbalanced after the last insertion? **ANS: 2**	1/2
3.	What is the time complexity of Heap Sort algorithm? **ANS: O(n log n)**	1/2
4.	 What is the post-order successor of node with value 5? **ANS: 2**	1/2

5.	Which of the following is/are correct preorder traversal sequence(s) of binary search tree(s)?
	`a. 4, 6, 7, 9, 18, 20, 19`
	`b. 3, 5, 7, 8, 15, 19, 25`
	`c. 2, 7, 10, 8, 14, 16, 20`
	`d. 5, 8, 9, 12, 10, 15, 25`
		1

	ANS: ALL	
6.	________________ data structure is used to implement Priority Queue. **ANS: heap**	1/2
7.	A red node in a Red-Black Tree can have which colour children? **ANS: black**	1/2
8.	How many minimum number of edges should a connected graph with n vertices have? ________ **ANS: n-1**	1/2
9.	If the adjacency matrix of a graph is symmetric then the graph is ____________ **ANS: undirected**	1/2
10.	Write the typedef for BST implemented using array. **ANS:** **typedef struct BST{** **int *A;** **int size;** **}BST;**	1/2
11.	Following nodes were inserted in an AVL tree in the given order: 200, 150, 300, 180, 190, 175. Node 190 was then deleted. Draw the resultant tree after deletion. **ANS:** **200, 175, 300, 150, 180** **(Level order) Diagrammatic representation is expected.**	1
12.	80, 90, 85, 100, 95, 87, 92 are the elements of a binary heap stored in an array. What is the content of the array after two consecutive delete operations? **ANS:** **87, 90, 92, 100, 95**	1

13. 80, 90, 85, 100, 95, 87, 92 were inserted in a Binary Search Tree in the same order. If the BST is represented using array, then element 100 will be inserted at _______________ index location of the array. — 1

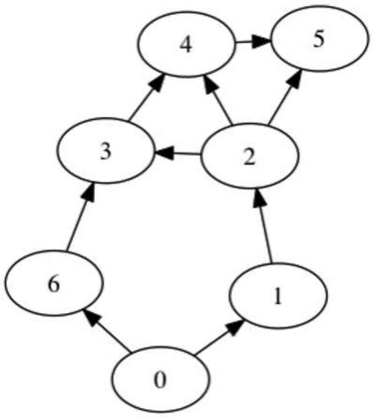
ANS: 7

14. After deleting an element from a max-heap, the maximum number of comparison operations required to re-heapify is 3. What is range of number of elements remaining in the heap now? — 1

__________________ to ________________

ANS: 5 - 7

15. Draw the Adjacency Matrix representation of the given graph. — 1



0	1	0	0	0	0	1
0	0	1	0	0	0	0
0	0	0	1	1	1	0
0	0	0	0	1	0	0
0	0	0	0	0	1	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0